NOVATECH BANK ROMANIA S.A.

CORPORATE AND METHODOLOGICAL MANUAL FOR LENDING TO INDIVIDUALS

Extended Methodological Rules, Risk Management Policies, Standard Operating Procedures (SOPs), and Prudential Compliance Guidelines

[Purpose: Evaluation and Indexing in Complex RAG Systems - Dissertation Project]

CHAPTER 1. INTRODUCTION AND INSTITUTIONAL REGULATORY FRAMEWORK

1.1. Structural Objectives of the Lending Manual

This manual establishes the comprehensive set of operational rules, internal control principles, and macroeconomic and microeconomic analysis criteria governing the activities of NovaTech Bank Romania S.A. The business objective is to provide a clear algorithmic and procedural structure capable of ensuring sustainable lending to individuals while maintaining the non-performing loan (NPL) ratio below a critical threshold of 2.1% at aggregate portfolio level.

This document serves as the single authoritative source for all departments involved in the loan lifecycle: the Retail Network Directorate, Risk Analysis Directorate, Anti-Fraud Department, Collateral Valuation Service, Compliance Department, and Workout/Debt Recovery Division.

1.2. Macroprudential Alignment with National Bank of Romania Regulations

Every directive mentioned in this manual is mapped directly to the prudential requirements of the National Bank of Romania (NBR). In particular, the calculation mechanisms in this document are adapted in accordance with NBR Regulation No. 17/2012 on lending conditions, as subsequently amended to prevent excessive household indebtedness. This manual incorporates the Basel III requirements on capital adequacy, liquidity risk, and financial leverage ratios.

The Bank applies a strict policy for monitoring the loan-to-value (LTV) ratio and maximum debt-to-income ratio (DTI) to ensure an optimal liquidity buffer should systemic shocks occur in Eastern European financial markets.

1.3. Periodic Updates and Reviews

The manual is reviewed semiannually or whenever the macroeconomic or legislative environment undergoes major changes. Any change to risk parameters, such as changing the FICO threshold or the haircut applied to income, must be approved by the Bank's Management Committee and endorsed by the Risk Management Directorate.

CHAPTER 2. CREDIT RISK GOVERNANCE PRINCIPLES AND DECISION-MAKING STRUCTURES

2.1. Risk Appetite Framework (RAF)

NovaTech Bank's risk appetite is defined annually by the Board of Directors. It establishes maximum loss-exposure limits, the optimal portfolio structure by product type, and client risk profiles. The Bank's strategy focuses on clients with stable, tax-deductible income and a clean record in credit databases.

The key structural performance indicators of the loan portfolio include Cost of Risk (CoR), Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). All these parameters are recalculated monthly using advanced statistical models.

2.2. Authority Matrix and Escalation Structures

The lending decision process is hierarchical to prevent the concentration of decision-making power and ensure independent dual control under the four-eyes principle:

- Level 1: Automated Decision (STP - Straight-Through Processing) - managed entirely by the risk engine for fully eligible applications.
- Level 2: Junior Retail Credit Analyst - authority limited to standard, exception-free loans of up to 50,000 RON.
- Level 3: Senior Credit Analyst / Supervisor - approval authority for amounts up to 100,000 RON or applications involving minor financial complexities.
- Level 4: Retail Risk Committee (CRR) - a collective body composed of risk and business directors, with full approval authority for loans of up to 150,000 RON, strategic exceptions, and applications in the Gray Zone.

CHAPTER 3. COMPLIANCE, ANTI-MONEY LAUNDERING (AML), AND KYC PROCEDURES

3.1. Know Your Customer (KYC) Standards in the Onboarding Process

Before initiating the financial creditworthiness analysis, the Bank is legally required to apply customer due diligence measures under Law No. 129/2019. This process involves fully identifying the applicant using official identity documents (identity card, passport), verifying the specimen signature, and collecting information on the purpose and nature of the business relationship and the source of the funds used for repayment.

The IT system performs real-time queries in the population records register (DEPABD) to validate the status of the identity document presented, eliminating the risk of using forged documents or documents reported lost.

3.2. Client Classification by AML Risk Profile

Each applicant is automatically assigned to an AML risk category: Low Risk, Standard Risk, or High Risk. Clients classified as high risk, such as those conducting large cash transactions or operating in sectors considered vulnerable, are subject to enhanced due diligence measures, and loan approval requires the Compliance Director's mandatory endorsement.

3.3. Treatment of Politically Exposed Persons (PEPs)

Politically Exposed Persons (PEPs), their family members, and persons known to be their close associates automatically receive a high-risk indicator in the system. The manual prohibits automatic loan approval for this category. PEP applications are escalated to the Risk Committee and require extremely rigorous documentation of the origin of the client's total wealth.

CHAPTER 4. ELIGIBILITY CRITERIA AND DETAILED APPLICANT PROFILE

4.1. Legal Status, Citizenship, and Tax Residence

The Bank grants credit facilities exclusively to individuals of legal age who have full legal capacity. From a legal and tax perspective, Romanian citizens and citizens of EU Member States are accepted if they hold a personal identification number (CNP) issued by the Romanian authorities, have permanent tax residence in Romania, and can be verified in the ANAF database.

For non-EU foreign citizens, lending is permitted only if they are married to a Romanian citizen, own real estate in Romania, and have held a local employment contract with the same employer for at least 24 months.

4.2. Age Limits and Their Relationship to Insurance Policies

The minimum accepted age is 21 on the application date. The applicant must not be older than 70 on the due date of the final installment. For clients over 62 at origination, the Bank requires a comprehensive life insurance policy covering death from any cause, with the insurance premium included in the annual percentage rate (DAE) calculation.

4.3. Analysis of Credit History in the Credit Bureau and CRB

A Credit Bureau query provides a clear picture of payment behavior over the past 48 months. Any record involving penalty collection, enforcement proceedings, repeated arrears, or fraud results in the application being immediately REJECTED.

Delinquency Filtering and Classification System:

- Delinquencies under 15 days: They are considered technical and do not affect eligibility, but may slightly reduce the internal score.
- Delinquencies from 16 to 30 days: They require written justification from the client and evidence that the situation has been remedied.
- Active delinquencies over 30 days: Automatically REJECTED.
- Historical delinquencies over 90 days in the past year: Automatically REJECTED, unless the debt has been fully settled and the client's current income is 50% higher than when the delinquency occurred (requires MANUAL REVIEW by the CRR).

CHAPTER 5. COMPREHENSIVE METHODOLOGY FOR INCOME ASSESSMENT AND WEIGHTING

5.1. Philosophy of Prudential Income Adjustment (Haircut Policy)

Applicants' declared income is subject to prudential adjustment coefficients known as 'haircuts'. The purpose of these weights is to protect the Bank against the volatility of non-salary income. While a fixed salary from the public sector or a multinational company is highly stable, income from self-employment, dividends, or rentals may fluctuate dramatically depending on the global economic environment.

5.2. Classification of Income Sources and Weighting Coefficients

The following structured table defines exactly how the IT system (Credit Engine) adjusts income before running the debt-to-income ratio (DTI) formulas:

Income Source Category	Internal Weight	Documentation and Validation Requirements
Salary (Permanent employment contract)	**100%**	Consent for an ANAF query. No physical documents.
Salary (Fixed-term employment contract)	**80%**	Contract for at least 24 months, with a renewal clause.
Old-age pension (Permanent)	**100%**	Final pension award decision + recent pension statement.
Self-employment/liberal professions income (PFA / PFI)	**75%**	Single Tax Return, tax assessment decisions for the last 2 cycles.
Dividends (Company shareholders)	**60%**	2 years of audited financial statements + a General Meeting of Shareholders (AGA) resolution approving dividend distribution.
Rental income (Real estate leases)	**50%**	ANAF-registered contract for at least 12 months + bank account statement.

Copyright / Intellectual Property Income	**70%**	Assignment agreements, 12-month history of recurring receipts.
Per Diem Allowances for Seafarers / Aircrew	**60%**	Valid seafarer employment / flight contract, bank statements in RON/foreign currency.
Management / Mandate Agreements	**85%**	Contract valid for at least 36 months, 6-month receipt history.

5.3. Filters for the Complete Exclusion of Unstable Income Sources

The Bank applies a highly conservative approach. The following sources receive a 0% weight and cannot be used for co-financing: income from meal vouchers, holiday vouchers, social assistance or temporary benefits, life annuities not registered for tax purposes, sports-betting winnings, speculative crypto-asset trading, or short-term equity trading (Day Trading).

CHAPTER 6. INTERNAL CREDIT SCORING MODEL AND SCORECARDS

6.1. Scorecard Algorithm Structure

NovaTech Bank's scoring system combines the client's demographic data (age, education level, marital status, housing status) with financial and behavioral data obtained from the Credit Bureau. Each indicator receives an internal score, and their sum generates the Internal Credit Score (SCI), scaled from 0 to 1000 points.

6.2. Correlation of the FICO Score with Bank Risk Bands

The external FICO score acts as an initial safety filter. The decision bands are:

- FICO 750 - 850 (Excellent Risk): The client receives instant STP processing and the most favorable interest-rate margins.
- FICO 650 - 749 (Standard Risk): Normal processing, standard income validation.
- FICO 620 - 649 (Substandard Risk / Gray Zone): The application is removed from automated processing and sent to a senior analyst for a thorough MANUAL REVIEW.
- FICO < 620 (Unacceptable Risk): Automatically REJECTED by the system.

CHAPTER 7. PRUDENTIAL LIMITS AND MAXIMUM DEBT-TO-INCOME RATIO (DTI)

7.1. Standard Calculation Formula under NBR Rules

The maximum debt-to-income ratio (DTI) is calculated strictly in accordance with the methodology imposed by the central bank, by dividing total monthly payment obligations by weighted eligible net income:

DTI (%) = [(Existing Loan Payments + New Loan Payment) / Weighted Eligible Net Income] × 100

7.2. Adjustments for Currency Risk and Interest-Rate Shocks

If a client requests a loan in EUR while earning income in RON, the Bank applies a currency stress test that simulates a sudden 15% depreciation of the Romanian leu. In addition, for variable-interest loans, a 2-percentage-point increase in the reference index (IRCC) is simulated. If DTI exceeds the legal threshold after these shocks are simulated, the loan is REJECTED.

CHAPTER 8. LOAN PRODUCT FEATURES AND PARAMETERS

8.1. Standard 'NovaFlex' Personal Loan

The core product offers financing between 5,000 and 150,000 RON for a fixed term of up to 60 months. Interest may be fixed for the entire term or variable (IRCC + the Bank's fixed margin). No application processing fees are charged to clients whose salary is paid into a NovaTech account.

8.2. 'Eco-Nevoi' Green Loan for Sustainability

Intended exclusively for environmental projects (photovoltaic panels, heat pumps, electric vehicles). Clients receive a special interest reduction of 1,500 basis points (1.5%), but must provide supporting documents (purchase invoices and acceptance reports) within 60 days of drawdown.

CHAPTER 9. CO-BORROWER STRUCTURE AND ANCILLARY INSURANCE

9.1. Rules on Co-Borrower Involvement

To increase the maximum amount that may be borrowed, the applicant may add one co-borrower (who must be a spouse or first-degree relative living at the same address). The co-borrower must meet the same tenure, eligibility, and history criteria as the primary applicant, and their combined income is added to the DTI calculation base.

9.2. Ancillary Insurance Policies (Unemployment, Life, Disability)

Insurance policies protect both the Bank and the client against unforeseen events. In the event of job loss, the insurer covers the monthly installments for up to 6 consecutive months.

CHAPTER 10. LOAN ORIGINATION AND DISBURSEMENT PROCESS FLOW

10.1. Stages of the Standard Operational Flow

The origination process consists of 4 major stages: Onboarding and KYC; Verification and Risk (ANAF/Credit Bureau query); Analysis and Decision (STP or MANUAL REVIEW); and Contract Signing and Funds Disbursement. All contractual documents are signed electronically using qualified digital certificates.

CHAPTER 11. POST-ORIGINATION MONITORING AND IFRS 9 PROVISIONING

11.1. Early Warning Indicators (Early Warning Signs - EWS)

The Bank runs early-warning algorithms weekly across active client accounts. Declining account turnover, new garnishments, or excessive Credit Bureau queries signal a deterioration in the risk profile, triggering preventive action by the Bank.

11.2. Loan Classification under IFRS 9

Loans are classified into 3 stages: Stage 1 (performing loans, normal risk), Stage 2 (a significant increase in credit risk - SICR, for example delinquencies over 30 days), and Stage 3 (non-performing/defaulted loans, delinquencies over 90 days). For each stage, the Bank sets aside specific provisions from its own capital.

CHAPTER 12. ARREARS MANAGEMENT, WORKOUT, AND RECOVERY

12.1. Soft Collection Flow (Days 1 - 30)

This process uses amicable customer outreach through automated digital channels, reminder calls, and short-term plans to clear negative balances.

12.2. Restructuring and Rescheduling Procedures

For clients experiencing hardship (unemployment, serious medical problems), the Bank may approve a loan restructuring: granting a 3-6 month grace period for principal payments or extending the loan term (without exceeding the legal limit of 5 years from the original date) to reduce the monthly payment.

12.3. Enforcement and Outsourcing of Receivables (Day 91+)

If the loan becomes more than 90 days past due and no restructuring agreement is reached, it is accelerated and becomes immediately due in full. The case is sent to partner judicial enforcement officers to obtain income garnishment orders or enforce against movable assets, or the receivable is sold to specialist debt-recovery companies.

ANNEXES AND TEST SCENARIOS FOR EVALUATING THE RAG ASSISTANT

Annex A: Synthetic Dataset (Complex Validation Cases)

To evaluate the RAG system's performance, the developer will use the following benchmark test cases to verify that the LLM correctly extracts rules and exceptions:

Test Case 1 (APPROVED with Exception E-3.1):

Applicant: IT programmer. Tenure with current employer: 4 months. Gap between jobs: 5 days. Previous tenure: 24 months. Net income: 15,000 RON. FICO: 720. Requests a loan with a monthly payment of 4,000 RON.

Correct RAG system answer: APPROVED. The client qualifies for exception E-3.1 concerning the IT sector (minimum accepted tenure of 3 months). DTI is 4,000 / 15,000 = 26.6%, below the 40% limit.

Test Case 2 (Automatically REJECTED Based on Age):

Applicant: Pensioner. Current age: 67. Eligible net income: 4,500 RON. FICO: 690. Requests a loan for a term of 5 years (60 months) with a payment of 1,200 RON.

Correct RAG system answer: REJECTED. At loan maturity, the client would be 67 + 5 = 72 years old, which violates the absolute maximum-age rule of 70 defined in Chapter 4.

Test Case 3 (Weighted Income Calculation - Haircut):

Applicant: Independent consultant (PFA). Average net income declared on the Single Tax Return: 10,000 RON per month. FICO: 680. Requests a loan with a payment of 3,500 RON.

Correct RAG system answer: REJECTED. Under the policies in Chapter 5, PFA income is assigned a 75% weighting under the haircut policy. The eligible income used in the calculation is 10,000 × 0.75 = 7,500 RON. The maximum permitted DTI for RON-denominated loans is 40%, which means a maximum payment of 7,500 × 0.40 = 3,000 RON. The requested payment of 3,500 RON exceeds the maximum permitted under prudential rules.

Annex B: Extended Standard Operating Procedures (SOP) Guide for MANUAL REVIEW

SOP Retail-Risk-01: Document Assessment in Hybrid Cases

This section sets out the methodological steps a second-line senior analyst must follow when reviewing a complex Group 1 application. The analyst compares 12 months of bank statements with the turnover reported to ANAF. If the amounts reported for tax purposes differ from those actually received by more than 10%, the prudential principle applies, and the lower amount is used to calculate repayment capacity.

The analyst must check the Trade Register and the court portal to determine whether the applicant's employer is subject to insolvency, judicial reorganization, or bankruptcy proceedings. If the employer has a high insolvency risk, the application is automatically REJECTED, regardless of the applicant's current creditworthiness.

SOP Retail-Risk-02: Document Assessment in Hybrid Cases

This section sets out the methodological steps a second-line senior analyst must follow when reviewing a complex Group 2 application. The analyst compares 12 months of bank statements with the turnover reported to ANAF. If the amounts reported for tax purposes differ from those actually received by more than 10%, the prudential principle applies, and the lower amount is used to calculate repayment capacity.

The analyst must check the Trade Register and the court portal to determine whether the applicant's employer is subject to insolvency, judicial reorganization, or bankruptcy proceedings. If the employer has a high insolvency risk, the application is automatically REJECTED, regardless of the applicant's current creditworthiness.

SOP Retail-Risk-03: Document Assessment in Hybrid Cases

This section sets out the methodological steps a second-line senior analyst must follow when reviewing a complex Group 3 application. The analyst compares 12 months of bank statements with the turnover reported to ANAF. If the amounts reported for tax purposes differ from those actually received by more than 10%, the prudential principle applies, and the lower amount is used to calculate repayment capacity.

The analyst must check the Trade Register and the court portal to determine whether the applicant's employer is subject to insolvency, judicial reorganization, or bankruptcy proceedings. If the employer has a high insolvency risk, the application is automatically REJECTED, regardless of the applicant's current creditworthiness.

SOP Retail-Risk-04: Document Assessment in Hybrid Cases

This section sets out the methodological steps a second-line senior analyst must follow when reviewing a complex Group 4 application. The analyst compares 12 months of bank statements with the turnover reported to ANAF. If the amounts reported for tax purposes differ from those actually received by more than 10%, the prudential principle applies, and the lower amount is used to calculate repayment capacity.

The analyst must check the Trade Register and the court portal to determine whether the applicant's employer is subject to insolvency, judicial reorganization, or bankruptcy proceedings. If the employer has a high insolvency risk, the application is automatically REJECTED, regardless of the applicant's current creditworthiness.

SOP Retail-Risk-05: Document Assessment in Hybrid Cases

This section sets out the methodological steps a second-line senior analyst must follow when reviewing a complex Group 5 application. The analyst compares 12 months of bank statements with the turnover reported to ANAF. If the amounts reported for tax purposes differ from those actually received by more than 10%, the prudential principle applies, and the lower amount is used to calculate repayment capacity.

The analyst must check the Trade Register and the court portal to determine whether the applicant's employer is subject to insolvency, judicial reorganization, or bankruptcy proceedings. If the employer has a high insolvency risk, the application is automatically REJECTED, regardless of the applicant's current creditworthiness.